Functional Equations Solutions

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§1 Walkthrough Problems

Problem 1.1 (USAMO 2002)**.** Determine all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(x^2 - y^2) = xf(x) - yf(y)$$

for all pairs of real numbers x and y.

Proof. Let $P(x, y)$ denote the given assertion. Note that $P(0, 0)$ gives us $f(0) = 0$. Then,

$$\begin{aligned} P(x, -x) \implies & f(x^2 - x^2) = xf(x) + xf(-x) \\ & 0 = x(f(x) + f(-x)) \end{aligned}$$

For nonzero x, we have $f(-x) = -f(x)$, but when $x = 0$ we have $f(0) = 0$ anyway, so we have f is odd for all real numbers x.

$$P(x, 0) \implies f(x^2) = xf(x)$$

So now we can rewrite the original condition as:

$$f(x^2 - y^2) = f(x^2) - f(y^2)$$

We let $a = x^2$, and $b = y^2$ such that, for nonnegative a, b we have:

$$f(a - b) = f(a) - f(b)$$

Again, we can set $a = x_0 + y_0$ and $b = y_0$ to get:

$$f(x_0 + y_0 - y_0) = f(x_0 + y_0) - f(y_0) \implies f(x_0 + y_0) = f(x_0) + f(y_0)$$

for nonnegative x_0, y_0. However, since f is odd, we get that f is additive on the negatives too.

Finally, we let $x = a + b$ in one of the previous equations $f(x^2) = xf(x)$ to get:

$$f(a^2 + 2ab + b^2) = (a + b)(f(a) + f(b))$$

$$f(2ab) = af(b) + bf(a)$$

$$a = 1 \implies f(2b) = f(b) + f(1)b = 2f(b)$$

Which gives $f(x) = cx$ for all x, which upon checking works. □

Problem 1.2 (IMO 2017)**.** Solve over $\mathbb{R}$ the functional equation

$$f(f(x)f(y)) + f(x+y) = f(xy).$$

Proof. We claim that $f(x) = 0$, $f(x) = 1 - x$, and $f(x) = x - 1$ are the only solutions. It's easy to check that these solutions indeed work.

Let $P(x, y)$ denote the given assertion. First, note:

$$P(0,0) \implies f(f(0)^2 = 0) \implies \text{ there exists a } z \text{ such that } f(z) = 0.$$

Also, $P(x, 0)$ gives that $f(0) = 0 \implies f \equiv 0$. So from now assume that $f(0) \neq 0$. We take, for $z \neq 1$,

$$P\left(z, \frac{z}{z-1}\right) \implies f\left(f(z)f\left(\frac{z}{z-1}\right)\right) = 0$$

giving $f(0) = 0$, which is a contradiction, giving $z = f(0)^2 = 1 \implies f(0) = \pm 1$. One can now take $P(1, x)$ to give $f(x+1) = f(x) - 1$, then $P(x, 0)$ to get $f(f(x)) = 1 - f(x)$. Applying f on this equation and equating some terms gives us:

$$f(f(f(x))) = f(1 - f(x)) = 1 - f(f(x)) = f(x)$$

Before we move on, I'd like to prove something that would help us in our final claim.

Claim 1.3 — For sufficiently large N_1 and N_2, we can find x, y satisfying $x + y = N_1 + 1$ and $xy = N_2$.

Proof. First we observe that we can think of x, y as roots of some quadratic equation. Of course, comparing it with Vieta's gives us that x and y are possible roots of the equation

$$x_0^2 - (N_1 + 1)x_0 + N_2 = 0$$

Obviously distinct solutions x, y to this equation exist if and only if the discriminant, Δ is greater than zero. So we have:

$$\begin{aligned} \Delta &= (N_1 + 1)^2 - 4N_2 \\ &= N_1^2 + 2N_1 + 1 - 4N_2 > 0 \end{aligned}$$

So we can just pick sufficiently large N_1, N_2 that satisfies the condition(or maybe something like $N_1^2 \geq 4N_2$) so we have solutions for x, y. □

Now, we prove the final claim.

Claim 1.4 — f is injective.

Proof. Assume for some sufficiently large a, b, we have $f(a) = f(b)$ and also, $x + y = a + 1$ and $xy = b$(from the above claim, we can do this). Then, the original assertion gives us

$$\begin{aligned} f(f(x)f(y)) &= f(b) - f(a+1) \\ &= f(b) - (f(a) - 1) \\ &= 1 \end{aligned}$$

Shifting f we get that $f(f(x)f(y) + 1) = 0 \implies f(x)f(y) + 1 = 1$ so $f(x)f(y) = 0$, giving that $1 \in \{x, y\}$. However, we are done as if $x = 1$ we get $y + 1 = a + 1 \implies y = a$ and $xy = b \implies b = y = a$. This follows similarly for $y = 1$, and so f is injective. □

From the injectivity, and our earlier calculations we get $f(1-f(x)) = f(x) \implies f(x) = 1-x$ and since $-f$ must be a solution too, $f(x) = x-1$ is another solution. Therefore, our solutions are:

$$\boxed{f(x) = 0, 1-x, x-1}$$

$\square$

Problem 1.5 (USAMO 2018)**.** Find all functions $f : (0, \infty) \to (0, \infty)$ such that

$$f\left(x + \frac{1}{y}\right) + f\left(y + \frac{1}{z}\right) + f\left(z + \frac{1}{x}\right) = 1$$

for all x, y, $z > 0$ with $xyz = 1$.

Proof. We begin by substituting the values $x = \frac{a}{b}$, $y = \frac{b}{c}$, $z = \frac{c}{a}$ to get the following:

$$\sum_{\text{cyc}} f\left(\frac{a+c}{b}\right) = 1$$

We then provide another substitution, this time for the function. Define a function $g : (0, 1) \to (0, 1)$ such that $g(x) = f(\frac{1-x}{x})$, then for $a + b + c = 1$ it follows that

$$g(a) + g(b) + g(c) = 1$$

Claim 1.6 — g satisfies Jensen's functional equation over $(0, \frac{1}{2})$.

Proof. Define variables x, $y \in (0, \frac{1}{2})$ such that $\frac{x+y}{2} \in (0, \frac{1}{2}) \subset (0, 1)$ and also $0 < x + y < 1$. Therefore, we can make the following substitutions:

$$a = b = \frac{x+y}{2}$$

$$c = 1 - (x + y)$$

Which satisfy $a + b + c = 1$, then,

$$g\left(\frac{x+y}{2}\right) + g\left(\frac{x+y}{2}\right) + g(1 - (x + y)) = 1$$

and

$$g\left(x\right) + g\left(y\right) + g(1 - (x + y)) = 1$$

Give us that

$$g(x) + g(y) = 2g\left(\frac{x+y}{2}\right),$$

as desired. $\square$

We now define a new function $h : [0, 1] \to \mathbb{R}$ as:

$$h(t) = g\left(\frac{2t+1}{8}\right) - (1-t)g\left(\frac{1}{8}\right) - tg\left(\frac{3}{8}\right)$$

Notice that $h(0) = h(1) = 0$. Also,

$$h\left(\frac{1}{2}\right) = g\left(\frac{2}{8}\right) - \frac{1}{2}g\left(\frac{1}{8}\right) - \frac{1}{2}\left(\frac{3}{8}\right)$$

$$= g\left(\frac{1}{8}\right) + g\left(\frac{1}{8}\right) - \frac{1}{2}g\left(\frac{1}{8}\right) - \frac{1}{2}g\left(\frac{1}{8}\right) - \frac{1}{2}g\left(\frac{1}{8}\right)$$
$$= 0$$

So now we have $h(0) = h(1) = h(\frac{1}{2}) = 0$. $\square$

§2 Problems

Problem 2.1 (IMO 2008)**.** Find all functions f from the positive reals to the positive reals such that

$$\frac{f(w)^2 + f(x)^2}{f(y^2) + f(z^2)} = \frac{w^2 + x^2}{y^2 + z^2}$$

for all positive real numbers w, x, y, z satisfying $wx = yz$.

Proof. We let $P(w, x, y, z)$ represent the given assertion. We claim that $f(x) = x$ and $f(x) = \frac{1}{x}$ are the only solutions, it is easy to check that these work.

First, note

$$P(x, x, x, x) \implies \frac{f(x)^2}{f(x^2)} = 1 \implies f(x^2) = f(x)^2$$

Which also gives $f(1) = 1$. Further,

$$P(x, 1, \sqrt{x}, \sqrt{x}) \implies (f(x) - x)(xf(x) - 1) = 0$$

Which gives us that $f(x) = x$ or $\frac{1}{x}$. However, as f could be a piecewise function, we prove the nonexistence of one.o

We assume $f(a) = a$ and $f(b) = \frac{1}{b}$ for some distinct a, b. We note:

$$P(\sqrt{a}, \sqrt{b}, \sqrt{ab}, 1) \implies \frac{a + \frac{1}{b}}{f(ab) + 1} = \frac{a + b}{ab + 1}$$

If we take $f(ab) = ab$, we get $b = 1$ and the case where $f(ab) = \frac{1}{ab}$ goves $a = 1$, contradicting that a and b are distinct. Thus, f is not piecewise, and the only solutions are $f(x) = x$ and $f(x) = \frac{1}{x}$ for all $x \in \mathbb{R}^+$. □

Problem 2.2 (IMO 2010)**.** Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that for all x, $y \in \mathbb{R}$,

$$f(\lfloor x \rfloor y) = f(x)\lfloor f(y) \rfloor$$

Proof. Let $P(x, y)$ denote the given assertion. Note that

$$P(0, 0) \implies f(0) = f(0)\lfloor f(0) \rfloor$$

From which we get that either $f(0) = 0$, or $1 \le f(0) < 2$, more specifically we get $\lfloor f(0) \rfloor = 1$. We now consider cases.

CASE 1. $f(0) = 0$ Let some x be such that $0 < x < 1$, then the original equation becomes:

$$f(x)\lfloor f(y) \rfloor = 0$$

Which gives us either $\lfloor f(y) \rfloor = 0$ or $f(x) = 0$ for all $0 < x < 1$. For the first case, consider now $P(1, y)$ giving us $f(y) = 0$ for all y. □

Problem 2.3 (IMO 2009)**.** Find all functions $f : \mathbb{Z}_{>0} \to \mathbb{Z}_{>0}$ such that for positive integers a and b, the numbers

$$a, \qquad f(b), \qquad , f(b + f(a) - 1)$$

are the sides of a non-degenerate triangle.

Proof. We prove something that will be crucial.

Claim 2.4 — If 1, a, b are sides of a triangle where a, $b \in \mathbb{Z}_{>0}$, then $a = b$.

Proof. By the triangle inequality,

$$b < a + 1$$

$$a < 1 + b$$

Combining gives us $-1 < a - b < 1$. Since a, b are positive integers, we get that $a - b = 0 \implies a = b$. □

Let $P(a, b)$ denote the tuple $(a, f(b), f(b + f(a) - 1))$. Then,

$$P(1, b) \implies (1, f(b), f(b + f(1) - 1))$$

From our above claim, we get that $f(b) = f(b + f(1) - 1)$, so $f(1) \neq 1 \implies f$ is periodic, and takes on finitely many values. Note that this also means f has a maximum(again, if and only if $f(1) \neq 1$).

Claim 2.5 — $f(1) = 1$.

Proof. We proceed by contradiction, assume not, then obviously $f(1) > 1$. We choose a sufficiently large a for $P(a, b)$, then by the triangle inequality,

$$a < f(b + f(a) - 1) - f(b)$$

Which would hold false. □

□